

Validity Analysis: FOLIO

Target context: US Undergraduate Logic Students — Introductory and Intermediate Courses

Overall risk: **HIGH**

Deployment Context

We are a US educational platform building a tool for American logic students. The tool should not only determine whether a logical argument is valid but also explain why — producing step-by-step reasoning traces that students can follow. We need to evaluate the LLM's ability to both perform logical reasoning and articulate its reasoning process in a pedagogically clear way.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ✓	3	Moderate gaps	high
Input Content	3	Moderate gaps	high
Input Form ✓	4	Minor gaps	high
Output Ontology ■	1	Serious concern	high
Output Content	2	Significant gaps	high
Output Form ■	1	Serious concern	high
Average	2.3		

■ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

FOLIO is a high-quality FOL deductive-reasoning benchmark with engine-verified labels, expert annotation, and topical diversity that partially serves a US undergraduate logic deployment — but only for one slice of what the deployment must evaluate. For the validity-precondition sub-judgment (does the conclusion follow from the premises?), FOLIO provides reliable ground truth and meaningful difficulty stratification, with strong support for the advanced predicate-logic register. However, the two HIGH-priority dimensions (output ontology and output form) are essentially completely misaligned: FOLIO outputs single labels, while the deployment requires numbered Fitch-style proof traces with rule citations, register-differentiated explanations, and countermodels for invalid arguments — none of which FOLIO produces, scores, or contains as ground truth. Secondary concerns include: no propositional-only coverage (deployment requires it), no

implicit-premise reconstruction (deployment requires it), notation mismatch between FOLIO's Russell & Norvig syntax and the LPL Fitch syntax students learn, and a ~10% FOL annotation-artifact rate observed in dataset sampling. P-FOLIO and emerging propositional-logic benchmarks (Rosetta-PL, DivLogicEval, the 2025 ITS study) partially fill some gaps but no off-the-shelf benchmark covers Fitch-style rule-cited proof traces or pedagogical countermodel construction.

Practical Guidance

What This Benchmark Measures

FOLIO can directly support evaluation of two sub-capabilities the deployment requires: (1) the binary/ternary deductive-validity judgment that must precede any proof generation (strongest dimension: input ontology and input form, with reliable output content for the label itself), and (2) NL-to-FOL translation correctness at a discourse-multi-premise level. For predicate-logic content at advanced register, FOLIO's depth-stratified evaluation provides genuine difficulty discrimination. It does not measure proof-trace generation, rule-citation correctness, register-appropriate explanation, or countermodel construction — the deployment's primary output requirements.

Construct Depth

Construct coverage is shallow relative to deployment needs: FOLIO probes the OUTCOME of deductive reasoning (final label) but not the STRUCTURE (numbered steps, rule applications, subproof discharges) or the PEDAGOGICAL FRAMING (register-appropriate prose, premise-citation correctness). The expert/non-expert 34-point performance gap [Q111] and the GPT-4-vs-expert 31.82-point gap [Q113] confirm FOLIO discriminates between sound and unsound FOL reasoning at the label level, but provides no evidence about explanation quality. The deepest gaps are in OO and OF (both HIGH-priority, both scored 1), with material secondary gaps in IO (propositional + implicit-premise) and OC (no explanation ground truth).

What Else You Need

Multi-source supplementation required: (1) For Fitch-style proof-trace evaluation — no off-the-shelf benchmark exists; build a custom rubric-based evaluation harness, optionally bootstrapped from P-FOLIO's NL proof chains [WEB-7] adapted to Fitch notation with rule labels; (2) For propositional logic — supplement with Rosetta-PL [WEB-11] or DivLogicEval [WEB-12], noting they also lack Fitch traces; (3) For countermodel evaluation — no benchmark covers this; develop with logic instructors using textbook countermodel exercises, drawing on the Lean-based 'Learning to Disprove' framework as a methodological template [WEB-10]; (4) For implicit-premise reconstruction — develop a custom enthymeme corpus from textbook exercises (Hurley, Bergmann/Moor/Nelson); (5) For pedagogical clarity / register appropriateness — adapt the 2025 ITS study's 4-criterion rubric [WEB-9] to Fitch-notation outputs. Additionally, clean ~10% of FOL annotation artifacts in FOLIO before any training use [DATASET-D24, D25, D26].

ID	Evidence
Q111	"Expert annotations achieve an accuracy of 95.98% while non-expert annotations achieves 61.82%, with a gap of 34.16%." (p.9)
Q113	"The expert and GPT-4 gap is 31.82%, suggesting significant room for model improvement." (p.9)
D24	premises-FOL contains: " $\neg(\exists y (\text{flannelShirt}(y) \wedge \text{WearEveryday}(x, y)) \wedge \text{Have}(\text{mike}, \text{highEnergy}) \wedge \text{Impulsive}(\text{mike})) \rightarrow (\text{Consistent}(\text{mike}) \wedge \text{StickTo}(\text{mike}, \text{theirRegularRoutine})) \oplus \neg \text{Like}(\text{mike}, \text{surprise})$ " — uses unbound variable x in outer formula — FOL annotation contains a free variable ' x ' not properly bound
D25	Lily is in James' family; she watches TV series in cinemas." — premises-FOL: " $\text{Customer}(\text{lily}) \wedge \text{In}(\text{lily}, \text{jameSFamily}) \wedge \text{WatchIn}(\text{lily}, \text{tv}, \text{cinema})$ " — Malformed FOL: missing closing parenthesis in the premises-FOL formula
D26	Matt does not invest in the public stock market regularly. Matt likes financial risks." — premises-FOL premise 6: " $\text{InvestInRegularly}(\text{matt}, \text{publicStockMarket})$ " — NL premise contradicts FOL annotation: NL says "does not invest" but FOL asserts $\text{InvestInRegularly}(\text{matt})$ without negation
WEB-7	P-FOLIO adds pass@k over multi-step NL proof chains — closest existing extension but still not Fitch-style rule-cited proofs (https://aclanthology.org/2024.findings-emnlp.966/)
WEB-9	2025 ITS study uses 4-criterion rubric (consistency, clarity, contextual explanation, pedagogical appropriateness) for LLM hints — closest to deployment evaluation but for propositional logic, not Fitch (https://arxiv.org/abs/2505.04736)
WEB-10	No benchmark evaluates formal countermodel construction as first-class output for pedagogical FOL/propositional logic (https://arxiv.org/pdf/2603.19514)

WEB-11	Rosetta-PL is a 2025 benchmark purpose-built for propositional logic LLM evaluation — confirms propositional-only coverage is a distinct need not addressed by FOLIO (https://arxiv.org/pdf/2505.00001)
WEB-12	https://arxiv.org/html/2509.15587v3

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: FOLIO | Context: US Undergraduate Logic Students — Introductory and Intermediate Courses

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

FOLIO's task taxonomy centers on FOL deductive reasoning with multi-premise discourse-level arguments [Q15, Q71, Q72] and an NL-FOL translation subtask [Q67], which partially align with the deployment's predicate-logic and NL-to-formal-form translation requirements. However, the benchmark explicitly excludes temporal and modal logic [Q139, Q140] (consistent with deployment scope) but also offers no propositional-only subtask — the dataset sample confirms every example uses FOL quantifiers even in minimal 2-premise cases [DATASET-D1, D31]. Argument reconstruction from arguments with implicit premises is also not present as a category; FOLIO annotators added commonsense premises explicitly [Q55]. Given MODERATE priority on IO, these omissions represent a moderate coverage gap relative to the deployment's tri-partite requirement (propositional, FOL, implicit-premise reconstruction).

ID	Evidence
Q15	"We present a natural language reasoning dataset, FOLIO, with first-order logic reasoning problems which require the models to decide the correctness of conclusions given a world defined by the premises." (p.1)
Q55	"Apart from checking whether NL and FOL express equivalent meanings, we also add necessary commonsense knowledge in both the NL and FOL premises. Sometimes humans do not write certain commonsense knowledge in the premises that is required in the FOL reasoning process, which is based solely on the premises given. We add such knowledge as additional premises at this stage." (p.5)
Q67	"We propose a new natural language to first-order logic translation (NL-FOL translation) task alongside our reasoning dataset." (p.6)
Q71	"Unlike previous work (Singh et al., 2020) which translates problems with a single premise and a single hypothesis, our task is on translating examples of various lengths with a focus on stories with multiple premises." (p.6)
Q72	"Thus, it also requires the models to consider discourse-level consistencies as opposed to translation at the sentence level." (p.6)
Q139	"Following (Russell and Norvig, 2010), we consider temporal logic and modal logic as special-purpose logics." (p.13)
Q140	"Consequently, they are beyond the scope of the definition of first-order logic used in our dataset." (p.13)
D1	No plants are fungi. Mushrooms are fungi. / $\forall x (\text{Plant}(x) \rightarrow \neg \text{Fungi}(x)) \quad \forall x (\text{Mushroom}(x) \rightarrow \text{Fungi}(x))$ — Simple 2-premise syllogism with parallel NL and FOL
D31	Each building is tall. Everything tall has height. — conclusion: "All buildings are magnificent. — Extremely short, minimal story (2 premises) — simple non-sequitur conclusion

Strengths

- Strong coverage of multi-step FOL deductive chains, with 28.7% of examples requiring 5+ reasoning depths [Q60], directly relevant to advanced-register predicate logic exercises
- Discourse-level multi-premise translation task [Q71, Q72] aligns with deployment's NL-to-formal-form translation need
- Syllogistic chains [Q43] map to canonical textbook exercise forms students encounter

ID	Evidence
Q43	"There are 256 logically distinct types of syllogisms and 24 of them are valid (Lehman, 1973). We use various combinations of 24 valid syllogisms. We also add in conjunction, disjunction, and implication." (p.4)
Q60	"As shown in Figure 1, the mode of reasoning depths is four in FOLIO. 28.7% of the examples need five or more depths of reasoning to infer the conclusions, while the previous datasets needed at most five reasoning depths as shown in Table 1." (p.5)

Q71	“Unlike previous work (Singh et al., 2020) which translates problems with a single premise and a single hypothesis, our task is on translating examples of various lengths with a focus on stories with multiple premises.” (p.6)
Q72	“Thus, it also requires the models to consider discourse-level consistencies as opposed to translation at the sentence level.” (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories per elicitation Q3/A3: propositional logic, predicate/FOL reasoning, and argument reconstruction with implicit premises from everyday/quasi-legal scenarios. Modal/temporal/inductive are out of scope.

IO-2: Omitted categories: (a) propositional-only reasoning — no examples in the sample use only sentential connectives without quantifiers [DATASET-D1, D31]; (b) implicit-premise/enthymeme reconstruction — annotators explicitly added commonsense premises [Q55], and dataset sampling confirms every premise needed is explicit [DATASET-D36, D19].

IO-3: Categories present that may consume evaluation weight without serving the deployment: heavy reliance on HybLogic template-derived syllogism combinations [Q42, Q43] and abstract fictional-world scenarios [DATASET-D9, D8] that do not reflect everyday/quasi-legal argument types the deployment targets.

IO-4: Content-validity gaps documented: (a) no propositional-only subtask; (b) no implicit-premise reconstruction subtask; (c) abstract template-derived scenarios consume evaluation weight beyond what deployment register requires. Modal/temporal exclusion [Q139, Q140] is aligned with deployment scope.

ID	Evidence
Q3	“In addition to the main NL reasoning task, NL-FOL pairs in FOLIO constitute a new NL-FOL translation dataset.” (p.1)
Q15	“We present a natural language reasoning dataset, FOLIO, with first-order logic reasoning problems which require the models to decide the correctness of conclusions given a world defined by the premises.” (p.1)
Q42	“In this approach, we first generate logically valid stories, which are templates containing abstract categories and entities, by combining multiple syllogisms into a single story template: the conclusion of one syllogism is used as a premise for the next syllogism.” (p.4)
Q43	“There are 256 logically distinct types of syllogisms and 24 of them are valid (Lehman, 1973). We use various combinations of 24 valid syllogisms. We also add in conjunction, disjunction, and implication.” (p.4)
Q55	“Apart from checking whether NL and FOL express equivalent meanings, we also add necessary commonsense knowledge in both the NL and FOL premises. Sometimes humans do not write certain commonsense knowledge in the premises that is required in the FOL reasoning process, which is based solely on the premises given. We add such knowledge as additional premises at this stage.” (p.5)
Q67	“We propose a new natural language to first-order logic translation (NL-FOL translation) task alongside our reasoning dataset.” (p.6)
Q139	“Following (Russell and Norvig, 2010), we consider temporal logic and modal logic as special-purpose logics.” (p.13)
Q140	“Consequently, they are beyond the scope of the definition of first-order logic used in our dataset.” (p.13)
D1	No plants are fungi. Mushrooms are fungi. / $\forall x (\text{Plant}(x) \rightarrow \neg \text{Fungi}(x)) \quad \forall x (\text{Mushroom}(x) \rightarrow \text{Fungi}(x))$ — Simple 2-premise syllogism with parallel NL and FOL
D8	Every person in Potterville that knows magic flies. All wizards in Potterville know magic. Harry, who lives in Potterville either yells or flies. — Fictional Potterville scenario referencing Harry/Potter (Harry Potter allusion)
D9	Everything in Size Town is big or small. All big things in Size Town are heavy... The bird is in Size Town and it is not both heavy and still.” — conclusion: “If the bird is small or still, then it is either unpredictable or changing. — Purely abstract Size Town scenario — no real-world content

D19	For a country, if effective monetary policy is possible, it must have successful inflation control and a strong national currency. There is an embargo on Russian foreign trade goods." — conclusion: "In Russia, an effective monetary policy is possible. — Real-world geopolitical/economic scenario involving Russia and embargos
D31	Each building is tall. Everything tall has height." — conclusion: "All buildings are magnificent. — Extremely short, minimal story (2 premises) — simple non-sequitur conclusion
D36	Everyone working at Meta has a high income. A person with a high income will not take a bus to their destination... James has a car or works at Meta. — Contemporary real-world scenario (Meta/Google) embedded in logic chain
WEB-11	Rosetta-PL is a 2025 benchmark purpose-built for propositional logic LLM evaluation — confirms propositional-only coverage is a distinct need not addressed by FOLIO (https://arxiv.org/pdf/2505.00001)
WEB-13	ProofWriter covers propositional-level chaining with limited vocabulary but no Fitch notation (https://aclanthology.org/2021.findings-acl.317.pdf)

Information Gaps

- No documented mapping between FOLIO's task taxonomy and Fitch-style natural deduction inference-rule taxonomy

Requires Expert Verification

- Whether the advanced-register subset of the curriculum specifically needs subproof-discharge categories ($\vee E$, $\neg I$, $\rightarrow I$ via subproof) that FOLIO's flat-FOL output cannot exercise

Remediation

Gap 1: No propositional-only subtask and no implicit-premise/enthymeme reconstruction subtask

Recommendation: Complement FOLIO with Rosetta-PL [WEB-11] or DivLogicEval [WEB-12] for propositional coverage; develop a custom enthymeme corpus drawn from US-textbook exercises (Hurley, Bergmann/Moor/Nelson) for implicit-premise reconstruction evaluation.

ID	Evidence
WEB-11	Rosetta-PL is a 2025 benchmark purpose-built for propositional logic LLM evaluation — confirms propositional-only coverage is a distinct need not addressed by FOLIO (https://arxiv.org/pdf/2505.00001)
WEB-12	https://arxiv.org/html/2509.15587v3

Input Content (IC)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: FOLIO | Context: US Undergraduate Logic Students — Introductory and Intermediate Courses

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

FOLIO's content is topically diverse (4,351 distinct words [Q62]; Wikipedia-seeded plus hybrid stories [Q33, Q58]) and explicitly bias-screened [Q47, Q127], which broadly fits a US undergraduate audience. Dataset sampling confirms wide topical coverage — science, sports, consumer goods, technology [DATASET-D4, D11, D21, D36]. However, FOLIO content is uniformly written as fully explicit, formalized premise sets [Q55], which mismatches the deployment's stated need for everyday/quasi-legal arguments with implicit premises. Several sampled examples involve geopolitically sensitive content (communist ideology, Russian sanctions, North Korean nationality) [DATASET-D6, D19, D16] that, while not pedagogically disqualifying, could create classroom-distraction concerns. With IC weighted MODERATE, the content profile partially serves the deployment but does not capture the implicit-premise everyday-argument register that students must learn to handle.

ID	Evidence
Q33	"We collected our dataset using two different methods in order to obtain examples that are both logically diverse and complex and have abundant abstract syntax tree (AST) variations." (p.4)
Q47	"Our dataset prioritizes realism and factual accuracy, steering clear of biases and stereotypes linked to identity markers like race, ethnicity, gender, sexuality, nationality, class, and religion." (p.4)
Q55	"Apart from checking whether NL and FOL express equivalent meanings, we also add necessary commonsense knowledge in both the NL and FOL premises. Sometimes humans do not write certain commonsense knowledge in the premises that is required in the FOL reasoning process, which is based solely on the premises given. We add such knowledge as additional premises at this stage." (p.5)
Q58	"Table 3 shows that examples based on Wikipedia make up the largest portion of FOLIO, with 304 stories, 1,353 NL and FOL premise pairs, and 753 NL and FOL conclusion pairs. Hybrid annotations consist of 183 stories with 1,054 NL and FOL premise pairs, and 682 NL and FOL conclusion pairs in total." (p.5)
Q62	"Table 3 shows that our dataset has a vocabulary of 4,351 words, and the examples based on Wikipedia account for 74% of the total vocabulary even though the WikiLogic stories take up only 63% of the total number of stories." (p.5)
Q127	"We took out stories that had strongly opinionated language and contained gender, racial, and classist biases." (p.13)
D4	No trick-shot artist in Yale's varsity team struggles with half court shots. Everyone on Yale's varsity team is someone who struggles with half court shots or who successfully shoots a high percentage of 3-pointers." — "Jack is bad at mid-range shots. — Basketball sports story requiring 4+ step chain
D6	All entrepreneurs at the business conference enjoy the opportunity of starting a business. Everyone at the business conference who is an ardent communist prefers state ownership of the means of production. Ho is at the business conference and prefers state ownership of the means of production. — Politically framed scenario with communist/investor dichotomy
D11	All velvet-finish lipsticks in the Rouge Dior set, Lunar New Year Limited Edition are refillable... ROUGE Dior Colored Lip Balm 999 is a lipstick in the set, and it either has 'rosewood' in its official description or has a velvet finish. — Real-world brand story about a specific luxury cosmetics product
D16	All PRC nationals are entitled to national social insurance coverage. Everyone in the Franco-China diplomatic conference is either a PRC national or a French national, but not both... Either Mei is a North Korean and can have medical bills partially covered, or neither is true. — Geopolitical scenario involving PRC, France, North Korea nationals
D19	For a country, if effective monetary policy is possible, it must have successful inflation control and a strong national currency. There is an embargo on Russian foreign trade goods." — conclusion: "In Russia, an effective monetary policy is possible. — Real-world geopolitical/economic scenario involving Russia and embargos

D21	All orphan planets are rogue planets. If PSO J318.5–22 is not both a rogue planet and a planet gravitationally bound by the Sun, then it is a rogue planet. — Scientific Wikipedia-seeded story about astronomy
D36	Everyone working at Meta has a high income. A person with a high income will not take a bus to their destination... James has a car or works at Meta. — Contemporary real-world scenario (Meta/Google) embedded in logic chain

Strengths

- Topical diversity from 304 Wikipedia-seeded stories yields 74% of vocabulary [Q62] — reduces topical overfitting risk
- Explicit bias-removal protocol [Q47, Q127] aligns with US classroom content-neutrality expectations
- Explicit-premise format directly matches textbook exercise style for the formal-exercise portion of the curriculum [DATASET-D20]

ID	Evidence
Q47	"Our dataset prioritizes realism and factual accuracy, steering clear of biases and stereotypes linked to identity markers like race, ethnicity, gender, sexuality, nationality, class, and religion." (p.4)
Q62	"Table 3 shows that our dataset has a vocabulary of 4,351 words, and the examples based on Wikipedia account for 74% of the total vocabulary even though the WikiLogic stories take up only 63% of the total number of stories." (p.5)
Q127	"We took out stories that had strongly opinionated language and contained gender, racial, and classist biases." (p.13)
D20	No battery-powered watch is automatic. All digital watches are battery-powered. Moonwatch is either a digital watch and an automatic, or it is neither. — Classic syllogistic chain about watches

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Most content does not require US-specific cultural knowledge; examples reference globally-recognizable entities (Olympic medals, universities, consumer brands) [DATASET-D13, D11]. Some Wikipedia-seeded stories use less familiar entities (Bobby Flynn, John Nash architect) [DATASET-D18, D12], but these do not block reasoning.

IC-2: Cultural alignment is broadly acceptable. Bias-removal protocol [Q127] reduces identity-linked stereotype risk. Several sampled examples touch politically sensitive geopolitical content (Russian sanctions [DATASET-D19], North Korean nationality [DATASET-D16], ardent communist [DATASET-D6]) — these are logically neutral but may cause classroom-context friction.

IC-3: Content does not require Western-specific knowledge in a disqualifying way; cross-cultural Wikipedia seeding [Q34] yields mixed-origin entities (Andy Chang from Hong Kong [DATASET-D7], PRC/French nationals [DATASET-D16]).

IC-4: INSUFFICIENT DOCUMENTATION — no regional annotators (US-undergrad-logic-instructor reviewers) have re-annotated FOLIO content for classroom appropriateness. Would need instructor panel review.

IC-5: Content gap: explicit-premise-only format does not exercise implicit-premise reconstruction [Q55, DATASET-D36]; abstract fictional scenarios (Size Town, Potterville) [DATASET-D9, D8] do not represent everyday/quasi-legal arguments the deployment targets.

ID	Evidence
----	----------

Q34	"WikiLogic: annotation from scratch using Wikipedia articles as seeds. At this annotation stage, the annotators are asked to select random Wikipedia pages by repeatedly using the Wikipedia Special Random link. The Wikipedia articles are used to develop ideas for topics to write new stories." (p.4)
Q55	"Apart from checking whether NL and FOL express equivalent meanings, we also add necessary commonsense knowledge in both the NL and FOL premises. Sometimes humans do not write certain commonsense knowledge in the premises that is required in the FOL reasoning process, which is based solely on the premises given. We add such knowledge as additional premises at this stage." (p.5)
Q62	"Table 3 shows that our dataset has a vocabulary of 4,351 words, and the examples based on Wikipedia account for 74% of the total vocabulary even though the WikiLogic stories take up only 63% of the total number of stories." (p.5)
Q127	"We took out stories that had strongly opinionated language and contained gender, racial, and classist biases." (p.13)
D6	All entrepreneurs at the business conference enjoy the opportunity of starting a business. Everyone at the business conference who is an ardent communist prefers state ownership of the means of production. Ho is at the business conference and prefers state ownership of the means of production. — Politically framed scenario with communist/investor dichotomy
D7	Andy Chang directed EndGame. Andy Chang is from Hong Kong." — conclusion: "All of Andy Chang's movies are filmed outside of Washington. — Wikipedia-seeded story about a real-world film director from Hong Kong
D8	Every person in Potterville that knows magic flies. All wizards in Potterville know magic. Harry, who lives in Potterville either yells or flies. — Fictional Potterville scenario referencing Harry/Potter (Harry Potter allusion)
D9	Everything in Size Town is big or small. All big things in Size Town are heavy... The bird is in Size Town and it is not both heavy and still." — conclusion: "If the bird is small or still, then it is either unpredictable or changing. — Purely abstract Size Town scenario — no real-world content
D11	All velvet-finish lipsticks in the Rouge Dior set, Lunar New Year Limited Edition are refillable... ROUGE Dior Colored Lip Balm 999 is a lipstick in the set, and it either has 'rosewood' in its official description or has a velvet finish. — Real-world brand story about a specific luxury cosmetics product
D12	John Nash designed Oxford Circus. John Nash is a British architect. Oxford Circus is the entrance to Oxford Circus tube station, a part of the Central line in 1900. — Wikipedia-seeded story about real architect John Nash
D13	All Olympic gold medal winners are professional athletes. All Nobel physics laureates are full-time scientists. Amy spends the most time on sports, or Amy is an Olympic gold medal winner. If Amy is not a Nobel physics laureate, then Amy is not an Olympic gold medal winner. — Multi-step chain combining disjunction, contrapositive, complex reasoning
D16	All PRC nationals are entitled to national social insurance coverage. Everyone in the Franco-China diplomatic conference is either a PRC national or a French national, but not both... Either Mei is a North Korean and can have medical bills partially covered, or neither is true. — Geopolitical scenario involving PRC, France, North Korea nationals
D18	Bobby Flynn finished 7th while competing on Australian Idol. Australian Idol competitors are Australian citizens." — conclusion: "Bobby Flynn flew to America in 2007. — Wikipedia-seeded story about Australian musician
D19	For a country, if effective monetary policy is possible, it must have successful inflation control and a strong national currency. There is an embargo on Russian foreign trade goods." — conclusion: "In Russia, an effective monetary policy is possible. — Real-world geopolitical/economic scenario involving Russia and embargos
D36	Everyone working at Meta has a high income. A person with a high income will not take a bus to their destination... James has a car or works at Meta. — Contemporary real-world scenario (Meta/Google) embedded in logic chain

Information Gaps

- No US-undergrad-instructor review of FOLIO content appropriateness for classroom use
- No data on whether geopolitically sensitive examples disproportionately affect specific demographic subgroups of students

Requires Expert Verification

- Whether examples involving communist/Russian/North Korean content should be filtered or contextualized for US classroom deployment

Remediation

Gap 1: Explicit-premise-only content; some geopolitically sensitive examples may distract in US classroom context

Recommendation: Filter or contextualize examples involving politically loaded predicates (communist, North Korean nationality, Russian sanctions [DATASET-D6, D16, D19]) for classroom deployment; develop an everyday/quasi-legal

enthymeme corpus to complement FOLIO's fully-explicit-premise content.

ID	Evidence
D6	All entrepreneurs at the business conference enjoy the opportunity of starting a business. Everyone at the business conference who is an ardent communist prefers state ownership of the means of production. Ho is at the business conference and prefers state ownership of the means of production. — Politically framed scenario with communist/investor dichotomy
D16	All PRC nationals are entitled to national social insurance coverage. Everyone in the Franco-China diplomatic conference is either a PRC national or a French national, but not both... Either Mei is a North Korean and can have medical bills partially covered, or neither is true. — Geopolitical scenario involving PRC, France, North Korea nationals
D19	For a country, if effective monetary policy is possible, it must have successful inflation control and a strong national currency. There is an embargo on Russian foreign trade goods." — conclusion: "In Russia, an effective monetary policy is possible. — Real-world geopolitical/economic scenario involving Russia and embargos

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: FOLIO | Context: US Undergraduate Logic Students — Introductory and Intermediate Courses

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input modality (text-only English [Q22]) and writing system (Latin + standard logical symbols) match the deployment exactly. Linguistic register is high-quality formal English with documented quality control (Grammarly check [Q49], English Literature reviewer pass [Q50], disambiguation conventions for disjunction/quantifiers/plurals [Q130-Q134]). One material concern remains: FOLIO uses Russell & Norvig FOL syntax [Q52], not the Fitch-style notation of the Barwise & Etchemendy curriculum [WEB-1, WEB-3]. The benchmark's own inference engine does not natively support its syntax and required a custom parser [Q149, Q150]. With IF priority LOWER and notation being the only meaningful concern (not modality), the dimension is well-aligned with minor reservations. Dataset sampling also identified some double-negative NL phrasing [DATASET-D33] that could create pedagogical noise.

ID	Evidence
Q22	"We use formal logic, i.e., FOL to ensure the logical validity of the examples written in NL and propose a new NL-FOL translation task." (p.2)
Q49	"Apart from grammar, we make sure the sentences in our dataset are highly natural. All the sentences are first checked with a grammar checking tool, Grammarly." (p.4)
Q50	"Our annotators who have graduated from or are senior students studying English Literature conducted a thorough round of review for grammatical correctness and language naturalness." (p.4)
Q52	"We adopt the FOL definitions and syntax most widely used in the AI community (Russell and Norvig, 2010)." (p.5)
Q130	"We always use "either-or" to express exclusive disjunction." (p.13)
Q131	"We use either "A or B" or "A or B, or both" to express inclusive disjunction." (p.13)
Q132	"It is more natural to say "Some A is B" rather than "there exists an A such that A is B."" (p.13)
Q133	""All A are B" can be more natural than "If A then B"." (p.13)
Q134	"Other common issues in NL quality include singular/plural issues, especially in statements that deal with both categories and individual members of those categories; as well as ambiguities resulting from improper introduction of, or failure to introduce, proper nouns." (p.13)
Q149	"The inference engine does not support input in the FOL syntax adopted by standard education material (Russell and Norvig, 2010), which is used in our dataset." (p.14)
Q150	"We therefore developed a FOL parser in order to convert the FOL formulas written by humans to the input format of the inference engine." (p.14)
D33	No satin-finish lipsticks in the set do not have 'rosewood' in its official description. — Double-negative NL phrasing ("No X do not have") that may create ambiguity
WEB-1	https://press.uchicago.edu/ucp/books/book/distributed/L/bo12734345.html
WEB-3	OpenLogicProject Fitch — LPL Fitch rule-name vocabulary ($\rightarrow$, $\wedge$, etc.) is distinct from FOLIO's flat-FOL representation (https://github.com/OpenLogicProject/fitch)

Strengths

- Text-only English-language input [Q22] matches deployment modality exactly
- Strong NL quality control: Grammarly check [Q49] plus English Literature reviewer pass [Q50]

- Explicit disambiguation conventions for disjunction, quantifier phrasing, singular/plural [Q130-Q134] reduce surface-form ambiguity
- Readability distribution documented [Q59, Q154] supports accessibility assessment

ID	Evidence
Q22	"We use formal logic, i.e., FOL to ensure the logical validity of the examples written in NL and propose a new NL-FOL translation task." (p.2)
Q49	"Apart from grammar, we make sure the sentences in our dataset are highly natural. All the sentences are first checked with a grammar checking tool, Grammarly." (p.4)
Q50	"Our annotators who have graduated from or are senior students studying English Literature conducted a thorough round of review for grammatical correctness and language naturalness." (p.4)
Q59	"We use the Dale-Chall Readability Formula (Dale and Chall, 1948, 1995) to show the text complexity of FOLIO following (Singh et al., 2023; Arps et al., 2022; Wei et al., 2021)." (p.5)
Q130	"We always use "either-or" to express exclusive disjunction." (p.13)
Q131	"We use either "A or B" or "A or B, or both" to express inclusive disjunction." (p.13)
Q132	"It is more natural to say "Some A is B" rather than "there exists an A such that A is B."" (p.13)
Q133	"All A are B" can be more natural than "If A then B"." (p.13)
Q134	"Other common issues in NL quality include singular/plural issues, especially in statements that deal with both categories and individual members of those categories; as well as ambiguities resulting from improper introduction of, or failure to introduce, proper nouns." (p.13)
Q154	"We show the distribution of readability in Figure 3." (p.14)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions match: both source and deployment are text-only English [Q22]. No image, audio, or multimodal mismatch. Dataset confirms uniformly text-formatted premises and conclusions [DATASET-D1 through D36].

IF-2: Regional infrastructure supports text input fully — ~66-72% of US undergraduates use laptop/desktop primary [WEB-5], appropriate for symbolic-input tasks. ~64% experience unstable internet [WEB-5], which is a delivery concern but not an input-form one.

IF-3: Domain-specific form difference: FOLIO uses Russell & Norvig flat FOL syntax [Q52] rather than Fitch-style numbered-line subproof notation [WEB-3, WEB-4]. This is a notation mismatch within the input-form layer that affects how formal-input strings are encoded.

IF-4: External-validity concerns: (a) FOL notation differs from Fitch syntax students are taught — no published mapping exists [WEB-3, WEB-14]; (b) some NL premises use double-negative phrasing that may be pedagogically noisy [DATASET-D33].

ID	Evidence
Q22	"We use formal logic, i.e., FOL to ensure the logical validity of the examples written in NL and propose a new NL-FOL translation task." (p.2)
Q52	"We adopt the FOL definitions and syntax most widely used in the AI community (Russell and Norvig, 2010)." (p.5)

Q149	"The inference engine does not support input in the FOL syntax adopted by standard education material (Russell and Norvig, 2010), which is used in our dataset." (p.14)
D1	No plants are fungi. Mushrooms are fungi. / $\forall x (\text{Plant}(x) \rightarrow \neg \text{Fungi}(x)) \quad \forall x (\text{Mushroom}(x) \rightarrow \text{Fungi}(x))$ — Simple 2-premise syllogism with parallel NL and FOL
D33	No satin-finish lipsticks in the set do not have 'rosewood' in its official description. — Double-negative NL phrasing ("No X do not have") that may create ambiguity
D36	Everyone working at Meta has a high income. A person with a high income will not take a bus to their destination... James has a car or works at Meta. — Contemporary real-world scenario (Meta/Google) embedded in logic chain
WEB-3	OpenLogicProject Fitch — LPL Fitch rule-name vocabulary ($\rightarrow E$, $\wedge I$, etc.) is distinct from FOLIO's flat-FOL representation (https://github.com/OpenLogicProject/fitch)
WEB-4	logicmatters.net documents that Fitch (LPL), Jaśkowski, Kalish-Montague, and Gamut natural deduction styles are notationally distinct (https://www.logicmatters.net/latex-for-logicians/nd/)
WEB-5	~66-72% of US undergraduates primarily use laptop/desktop — supports text/symbolic input (https://www.newamerica.org/education-policy/edcentral/how-do-community-college-students-access-online-learning/)
WEB-14	el-sambal/fitch-proof documents LPL rule syntax with no cross-reference to FOLIO's operators (https://github.com/el-sambal/fitch-proof)

Information Gaps

- No published systematic mapping between FOLIO's Russell & Norvig operators and LPL Fitch rule names

Remediation

Gap 1: Russell & Norvig FOL syntax mismatches Fitch-style notation students learn

Recommendation: Develop and publish a systematic mapping between FOLIO's flat-FOL operators and LPL Fitch rule names ($\rightarrow E$, $\wedge I/\wedge E$, $\vee E$ with subproof, $\forall E$, $\exists E$ with subproof, $\neg I$); incorporate the `lplfitch` LaTeX package and `el-sambal/fitch-proof` checker [WEB-14] as reference encoders.

ID	Evidence
WEB-14	el-sambal/fitch-proof documents LPL rule syntax with no cross-reference to FOLIO's operators (https://github.com/el-sambal/fitch-proof)

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: FOLIO | Context: US Undergraduate Logic Students — Introductory and Intermediate Courses

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

FOLIO's output ontology is a three-way truth-value label (True/False/Unknown) determined by a FOL inference engine [Q36, Q66, Q94]. The deployment requires fundamentally different outputs: numbered Fitch-style proof traces with premise citations and inference-rule names, plus countermodels for invalid arguments (elicitation A1, A2). Dataset sampling confirms the schema contains exactly six fields with no proof-trace, rule-citation, or countermodel artifact for any example [DATASET-D1, D14, D4]. With OO priority HIGH, the output-ontology mismatch is essentially total: FOLIO labels every False example with only 'False', providing no countermodel for the deployment's required artifact [DATASET-D14]. The operator inventory [Q138] overlaps with Fitch connectives but no rule-vocabulary mapping exists [WEB-14].

ID	Evidence
Q36	"Each of the stories is composed of several premises and conclusions with truth values of True, False, or Unknown (see Table 2 for an example)." (p.4)
Q66	"Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning." (p.6)
Q94	"The majority baseline of our dataset is 38.5% since in our test set, there are 87, 78 and 61 examples with labels of true, false and unknown respectively." (p.7)
Q138	"We include the following operators: negation $\neg$, conjunction $\wedge$, disjunction $\vee$, implication $\rightarrow$, universal quantifier $\forall$, existential quantifier $\exists$, equal $=$." (p.13)
D1	No plants are fungi. Mushrooms are fungi. / $\forall x (\text{Plant}(x) \rightarrow \neg \text{Fungi}(x)) \forall x (\text{Mushroom}(x) \rightarrow \text{Fungi}(x))$ — Simple 2-premise syllogism with parallel NL and FOL
D4	No trick-shot artist in Yale's varsity team struggles with half court shots. Everyone on Yale's varsity team is someone who struggles with half court shots or who successfully shoots a high percentage of 3-pointers." — "Jack is bad at mid-range shots. — Basketball sports story requiring 4+ step chain
D14	Michael O'Donnell is a British physician, journalist, author, and broadcaster... Michael O'Donnell was born in Yorkshire" — conclusion: "There are no journalists that were born in Yorkshire. — Simple false universal overgeneralization from a positive instance
WEB-14	el-sambal/fitch-proof documents LPL rule syntax with no cross-reference to FOLIO's operators (https://github.com/el-sambal/fitch-proof)

Strengths

- Engine-verified three-way labels [Q36, Q66] are reliable ground truth for the deductive-validity sub-judgment the deployment must make before producing a proof
- Three-way distinction (True/False/Unknown) [Q94] identifies two distinct invalid-argument modes (contradiction vs. underdetermination) — useful signal even if countermodels are absent
- Standard FOL operator inventory [Q138] overlaps with Fitch connective set

ID	Evidence
Q36	"Each of the stories is composed of several premises and conclusions with truth values of True, False, or Unknown (see Table 2 for an example)." (p.4)

Q66	"Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning." (p.6)
Q94	"The majority baseline of our dataset is 38.5% since in our test set, there are 87, 78 and 61 examples with labels of true, false and unknown respectively." (p.7)
Q138	"We include the following operators: negation $\neg$, conjunction $\wedge$, disjunction $\vee$, implication $\rightarrow$, universal quantifier $\forall$, existential quantifier $\exists$, equal $=$." (p.13)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output categories required by deployment: (a) validity judgment (True/False/Unknown is approximately compatible); (b) numbered proof-trace with rule citations (absent); (c) countermodel for invalid arguments (absent); (d) register-differentiated explanation form (absent).

OO-2: Missing categories specific to deployment: countermodel construction for False-labeled examples [Q66, DATASET-D14]; proof-trace structure (numbered lines, subproof boxes, rule-citation field) — none present in the FOLIO schema [DATASET schema observation in Concern 1]; register tags for introductory vs. advanced (absent).

OO-3: FOLIO encodes Russell & Norvig FOL conventions [Q52] rather than Fitch-style rule taxonomy used by the LPL curriculum [WEB-3, WEB-4]. The decision rule (engine-verified single label) is curriculum-neutral but ignores the pedagogical structure of proof.

OO-4: Stakeholder-driven redesign is warranted: a Fitch-rule-aligned proof-trace ontology would require introducing per-step rule labels, premise-citation fields, and subproof-structure tags — categories entirely absent from FOLIO.

OO-5: Structural-validity violation: the construct 'logical argument validity in pedagogical context' has substantive internal structure (rule application, subproof discharge, countermodel) that FOLIO's single-label ontology collapses into a binary/ternary judgment. Content-validity violation: countermodel and rule-citation categories are entirely missing.

ID	Evidence
Q36	"Each of the stories is composed of several premises and conclusions with truth values of True, False, or Unknown (see Table 2 for an example)." (p.4)
Q52	"We adopt the FOL definitions and syntax most widely used in the AI community (Russell and Norvig, 2010)." (p.5)
Q66	"Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning." (p.6)
Q94	"The majority baseline of our dataset is 38.5% since in our test set, there are 87, 78 and 61 examples with labels of true, false and unknown respectively." (p.7)
Q138	"We include the following operators: negation $\neg$, conjunction $\wedge$, disjunction $\vee$, implication $\rightarrow$, universal quantifier $\forall$, existential quantifier $\exists$, equal $=$." (p.13)
D14	Michael O'Donnell is a British physician, journalist, author, and broadcaster... Michael O'Donnell was born in Yorkshire" — conclusion: "There are no journalists that were born in Yorkshire. — Simple false universal overgeneralization from a positive instance
WEB-3	OpenLogicProject Fitch — LPL Fitch rule-name vocabulary ($\rightarrow$ E, $\wedge$ I, etc.) is distinct from FOLIO's flat-FOL representation (https://github.com/OpenLogicProject/fitch)

WEB-4	logicmatters.net documents that Fitch (LPL), Jaśkowski, Kalish-Montague, and Gamut natural deduction styles are notationally distinct (https://www.logicmatters.net/latex-for-logicians/nd/)
WEB-10	No benchmark evaluates formal countermodel construction as first-class output for pedagogical FOL/propositional logic (https://arxiv.org/pdf/2603.19514)
WEB-14	el-sambal/fitch-proof documents LPL rule syntax with no cross-reference to FOLIO's operators (https://github.com/el-sambal/fitch-proof)

Information Gaps

- Whether the inference engine, internally, generates countermodels as a byproduct for False-labeled examples (not exposed in schema)

Requires Expert Verification

- Whether a custom Fitch-rule-tagged extension of FOLIO is feasible as a supplementation strategy

Remediation

Gap 1: No proof-trace structure or countermodel artifacts in output ontology; FOLIO collapses to single label

Recommendation: Build a Fitch-aligned output schema (numbered lines, rule-name field, premise-citation field, subproof-box markers, countermodel-as-domain-assignment field) and elicit reference proof traces from logic-instructor annotators for a curated subset of FOLIO stories; use P-FOLIO [WEB-7] as a starting NL-chain corpus and convert to Fitch notation.

ID	Evidence
WEB-7	P-FOLIO adds pass@k over multi-step NL proof chains — closest existing extension but still not Fitch-style rule-cited proofs (https://aclanthology.org/2024.findings-emnlp.966/)

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: FOLIO | Context: US Undergraduate Logic Students — Introductory and Intermediate Courses

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

For the labels FOLIO does provide, ground-truth quality is high: engine-verified [Q2, Q57], multi-stage expert annotation [Q29, Q124, Q125], 39.2% rewrite rate to address quality issues [Q48], and expert-annotator accuracy of 95.98% [Q111] confirming label reliability. However, with OC priority MODERATE, two distinct concerns lower the score substantially: (1) no ground-truth content exists for pedagogically-acceptable explanations, Fitch-style rule citations, register-appropriate prose, or countermodels — the artifacts the deployment needs to evaluate; (2) dataset sampling found ~10% rate of FOL annotation artifacts (free variables, missing parentheses, NL/FOL contradiction, name mismatches) [DATASET-D24, D25, D26, D23, D30, D27] that could propagate as incorrect ground truth if used for NL-FOL translation evaluation or model training.

ID	Evidence
Q2	"The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine." (p.1)
Q29	"At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved. For the FOL quality check, only annotators who are experts in first-order logic are involved." (p.4)
Q48	"Toward these objectives, we manually screened all stories and found that 39.2% of the stories suffer from at least one of these issues. We implemented a detailed protocol to rewrite these stories." (p.4)
Q57	"Recognizing that the FOL formula annotations can be error-prone, we verify the syntactic validity and label consistency of FOL formula annotations with an FOL inference engine." (p.5)
Q111	"Expert annotations achieve an accuracy of 95.98% while non-expert annotations achieves 61.82%, with a gap of 34.16%." (p.9)
Q124	"At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved." (p.12)
Q125	"For the FOL quality check, only annotators who are experts in first-order logic are involved." (p.12)
D23	No athletes never exercise. All professional basketball players are athletes... Either John is a professional basketball player and he never exercises, or he is not a professional basketball player and he sometimes exercises." — NOTE: FOL annotation says "jim" while NL says "John" — Annotation inconsistency: NL uses "John," FOL uses "jim"
D24	premises-FOL contains: " $\neg(\exists y (\text{flannelShirt}(y) \wedge \text{WearEveryday}(x, y)) \wedge \text{Have}(\text{mike}, \text{highEnergy}) \wedge \text{Impulsive}(\text{mike})) \rightarrow (\text{Consistent}(\text{mike}) \wedge \text{StickTo}(\text{mike}, \text{theirRegularRoutine})) \oplus \neg\text{Like}(\text{mike}, \text{surprise})$ " — uses unbound variable x in outer formula — FOL annotation contains a free variable 'x' not properly bound
D25	Lily is in James' family; she watches TV series in cinemas." — premises-FOL: " $\text{Customer}(\text{lily}) \wedge \text{In}(\text{lily}, \text{jameSFamily}) \wedge \text{WatchIn}(\text{lily}, \text{TV}, \text{cinema})$ " — Malformed FOL: missing closing parenthesis in the premises-FOL formula
D26	Matt does not invest in the public stock market regularly. Matt likes financial risks." — premises-FOL premise 6: " $\text{InvestInRegularly}(\text{matt}, \text{publicStockMarket})$ " — NL premise contradicts FOL annotation: NL says "does not invest" but FOL asserts $\text{InvestInRegularly}(\text{matt})$ without negation
D27	"($\text{Knows}(\text{dan}, \text{dune}) \wedge \text{ScienceFiction}(\text{dune}) \vee \text{ProvedToBe}(\text{dune}, \text{false})$)" — extra closing parenthesis in last premise FOL — Minor FOL syntax artifact (extra parenthesis)
D30	"All players must reach the yellow stage before they can reach the green stage. The yellow stage comes after the red stage." — premises-FOL contains " $\exists x \exists y \exists y \exists w$ " (duplicate variable y) — Minor FOL annotation typo: variable y declared twice

Strengths

- Engine-verified labels [Q2, Q57] provide deterministic ground truth for deductive validity
- Two-stage expert quality control (NLP/CL experts for NL, FOL experts for FOL) [Q29, Q124, Q125]
- 39.2% of stories rewritten to address quality issues [Q48] — substantial QC investment
- Expert annotators achieve 95.98% accuracy [Q111], confirming labels are well-formed

ID	Evidence
Q2	"The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine." (p.1)
Q29	"At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved. For the FOL quality check, only annotators who are experts in first-order logic are involved." (p.4)
Q48	"Toward these objectives, we manually screened all stories and found that 39.2% of the stories suffer from at least one of these issues. We implemented a detailed protocol to rewrite these stories." (p.4)
Q57	"Recognizing that the FOL formula annotations can be error-prone, we verify the syntactic validity and label consistency of FOL formula annotations with an FOL inference engine." (p.5)
Q111	"Expert annotations achieve an accuracy of 95.98% while non-expert annotations achieves 61.82%, with a gap of 34.16%." (p.9)
Q124	"At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved." (p.12)
Q125	"For the FOL quality check, only annotators who are experts in first-order logic are involved." (p.12)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Truth-value labels are derived from formal inference and are not subject to regional-stakeholder disagreement — formal validity is invariant across US logic curricula. However, the deployment's required outputs (proof traces, countermodels, register-appropriate explanations) have no ground truth in FOLIO at all.

OC-2: No disagreement expected on True/False/Unknown labels because they are engine-verified [Q2]. Disagreement is irrelevant because the deployment-relevant output types are absent.

OC-3: Annotator demographics are documented at a high level: college/graduate students, native or near-native English, formal FOL coursework [Q27, Q28, Q122, Q123]. No fine-grained demographic breakdown is provided; geographic distribution of annotators unspecified.

OC-4: Re-annotation by US-logic-instructor pool would be needed to introduce ground-truth Fitch-style proof traces, rule citations, register-tagged explanations, and countermodels — none exist in FOLIO. Dataset sampling also revealed FOL annotation errors that warrant cleanup [DATASET-D24, D25, D26].

OC-5: Aggregation method (single engine-verified label per example) cannot encode minority perspectives because formal validity admits no minority view. The relevant concern is not aggregation but absence of explanation-quality labels.

OC-6: Convergent-validity concern is minimal for the label that is present; external-validity concern is high because no labels exist for the explanation/countermodel artifacts the deployment must evaluate. Additionally, ~10% FOL-annotation-artifact rate observed in sample [DATASET-D24, D25, D26, D23, D30, D27] could harm convergent validity

for NL-FOL translation evaluation.

ID	Evidence
Q2	"The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine." (p.1)
Q27	"Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English." (p.3)
Q28	"They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or seman-" (p.3)
Q53	"In preliminary investigations, we found that the human-written FOL formulas suffer from FOL consistency issues, which necessitates an additional round of quality control for FOL formulas." (p.5)
Q57	"Recognizing that the FOL formula annotations can be error-prone, we verify the syntactic validity and label consistency of FOL formula annotations with an FOL inference engine." (p.5)
Q111	"Expert annotations achieve an accuracy of 95.98% while non-expert annotations achieves 61.82%, with a gap of 34.16%." (p.9)
Q122	"Given the complexities of our annotations, we selected annotators based on a few important criteria 1). Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English." (p.12)
Q123	"They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or semantic parsing." (p.12)
D23	No athletes never exercise. All professional basketball players are athletes... Either John is a professional basketball player and he never exercises, or he is not a professional basketball player and he sometimes exercises." — NOTE: FOL annotation says "jim" while NL says "John" — Annotation inconsistency: NL uses "John," FOL uses "jim"
D24	premises-FOL contains: " $\neg(\exists y (\text{flannelShirt}(y) \wedge \text{WearEveryday}(x, y)) \wedge \text{Have}(\text{mike}, \text{highEnergy}) \wedge \text{Impulsive}(\text{mike})) \rightarrow (\text{Consistent}(\text{mike}) \wedge \text{StickTo}(\text{mike}, \text{theirRegularRoutine})) \oplus \neg \text{Like}(\text{mike}, \text{surprise})$ " — uses unbound variable x in outer formula — FOL annotation contains a free variable 'x' not properly bound
D25	Lily is in James' family; she watches TV series in cinemas." — premises-FOL: " $\text{Customer}(\text{lily}) \wedge \text{In}(\text{lily}, \text{jameSFamily}) \wedge \text{WatchIn}(\text{lily}, \text{TV}, \text{cinema})$ " — Malformed FOL: missing closing parenthesis in the premises-FOL formula
D26	Matt does not invest in the public stock market regularly. Matt likes financial risks." — premises-FOL premise 6: " $\text{InvestInRegularly}(\text{matt}, \text{publicStockMarket})$ " — NL premise contradicts FOL annotation: NL says "does not invest" but FOL asserts $\text{InvestInRegularly}(\text{matt})$ without negation
D27	" $(\text{Knows}(\text{dan}, \text{dune}) \wedge \text{ScienceFiction}(\text{dune})) \vee \text{ProvedToBe}(\text{dune}, \text{false})$ " — extra closing parenthesis in last premise FOL — Minor FOL syntax artifact (extra parenthesis)
D30	"All players must reach the yellow stage before they can reach the green stage. The yellow stage comes after the red stage." — premises-FOL contains " $\exists x \exists y \exists y \exists w$ " (duplicate variable y) — Minor FOL annotation typo: variable y declared twice
WEB-7	P-FOLIO adds pass@k over multi-step NL proof chains — closest existing extension but still not Fitch-style rule-cited proofs (https://aclanthology.org/2024.findings-emnlp.966/)

Information Gaps

- Full FOL-annotation error rate across all 1,430 examples (sample-based lower-bound is ~10%)
- Fine-grained annotator demographics beyond 'college/graduate students with FOL coursework'

Requires Expert Verification

- Whether observed FOL annotation artifacts are systematic or distributed; would benefit from re-running the inference engine across the full corpus

Remediation

Gap 1: ~10% FOL annotation-artifact rate observed in dataset sample (free variables, missing parens, NL/FOL contradiction, name mismatches)

Recommendation: Re-run the inference engine on the full corpus and audit examples flagged in the dataset analysis [DATASET-D24, D25, D26, D23, D30, D27]; produce a cleaned subset before using FOLIO for any training or NL-FOL translation evaluation.

ID	Evidence
D23	No athletes never exercise. All professional basketball players are athletes... Either John is a professional basketball player and he never exercises, or he is not a professional basketball player and he sometimes exercises." — NOTE: FOL annotation says "jim" while NL says "John" — Annotation inconsistency: NL uses "John," FOL uses "jim"
D24	premises-FOL contains: " $\neg(\exists y (\text{flannelShirt}(y) \wedge \text{WearEveryday}(x, y)) \wedge \text{Have}(\text{mike}, \text{highEnergy}) \wedge \text{Impulsive}(\text{mike})) \rightarrow (\text{Consistent}(\text{mike}) \wedge \text{StickTo}(\text{mike}, \text{theirRegularRoutine})) \oplus \neg\text{Like}(\text{mike}, \text{surprise})$ " — uses unbound variable x in outer formula — FOL annotation contains a free variable 'x' not properly bound
D25	Lily is in James' family; she watches TV series in cinemas." — premises-FOL: " $\text{Customer}(\text{lily}) \wedge \text{In}(\text{lily}, \text{jameSFamily}) \wedge \text{WatchIn}(\text{lily}, \text{tv}, \text{cinema})$ " — Malformed FOL: missing closing parenthesis in the premises-FOL formula
D26	Matt does not invest in the public stock market regularly. Matt likes financial risks." — premises-FOL premise 6: " $\text{InvestInRegularly}(\text{matt}, \text{publicStockMarket})$ " — NL premise contradicts FOL annotation: NL says "does not invest" but FOL asserts $\text{InvestInRegularly}(\text{matt})$ without negation
D27	"($\text{Knows}(\text{dan}, \text{dune}) \wedge \text{ScienceFiction}(\text{dune}) \vee \text{ProvedToBe}(\text{dune}, \text{false})$)" — extra closing parenthesis in last premise FOL — Minor FOL syntax artifact (extra parenthesis)
D30	"All players must reach the yellow stage before they can reach the green stage. The yellow stage comes after the red stage." — premises-FOL contains " $\exists x \exists y \exists y \exists w$ " (duplicate variable y) — Minor FOL annotation typo: variable y declared twice

Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: FOLIO | Context: US Undergraduate Logic Students — Introductory and Intermediate Courses

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

FOLIO's output form is single-label classification accuracy [Q80] plus, for the translation subtask, Syntactic Validity and Execution Accuracy [Q73-Q76]. The deployment requires evaluation of structured, register-differentiated, rule-justified multi-step proof traces and countermodels — none of which FOLIO's metrics assess. With OF priority HIGH, the output-form mismatch is essentially complete. The evaluation infrastructure (Stanford CS221 inference engine + custom Python parser [Q148-Q151]) is purpose-built for label verification, and the authors themselves flag the need for a 'more reliable metric of NL-FOL translation' [Q77]. Dataset sampling confirms no proof-trace or countermodel artifact exists at the data level [Concern 1]. The only sub-task FOLIO can directly score for the deployment is the binary validity precondition, which is necessary but radically insufficient.

ID	Evidence
Q73	"Two metrics are adopted to evaluate NL-FOL translation to capture different aspects of the generation results: 1). Syntactic validity (SynV)." (p.6)
Q74	"The Syntactic Validity score measures whether the FOL formulas are syntactically valid. The score will be 1 if all FOL formulas of an example can pass the syntactic check and 0 otherwise 2)." (p.6)
Q75	"Inference Engine execution accuracy (ExcAcc). The group of translated FOL for premises and conclusions in one story is fed into our inference engine to output the truth value for each conclusion." (p.6)
Q76	"We define the accuracy of the output labels as the execution accuracy." (p.6)
Q77	"We leave for future work the design of a more reliable metric of NL-FOL translation." (p.6)
Q80	"We use accuracy for evaluating logical reasoning performance." (p.6)
Q148	"Although there are many provers widely used in the community (McCune, 2005–2010; Sutcliffe, 2017; Nipkow et al., 2002), we adopt the inference engine provided in the Stanford CS221 course page, which is a compact module designed specifically for procesing first-order logic statements." (p.14)
Q149	"The inference engine does not support input in the FOL syntax adopted by standard education material (Russell and Norvig, 2010), which is used in our dataset." (p.14)
Q150	"We therefore developed a FOL parser in order to convert the FOL formulas written by humans to the input format of the inference engine." (p.14)
Q151	"The converter is a semantic parser tool written in Python." (p.14)

Strengths

- Accuracy metric [Q80] is appropriate for the deductive-validity sub-judgment that must occur before proof generation
- Translation Execution Accuracy [Q75, Q76] measures whether translated FOL preserves semantics — partially relevant to NL-to-formal-form correctness in the deployment
- Confusion-matrix and depth-stratified analysis [Q99, Q155-Q157] reveal failure modes (over-prediction of True; harder at depths 4-7) that inform deployment risk assessment

ID	Evidence
Q75	"Inference Engine execution accuracy (ExcAcc). The group of translated FOL for premises and conclusions in one story is fed into our inference engine to output the truth value for each conclusion." (p.6)

Q76	"We define the accuracy of the output labels as the execution accuracy." (p.6)
Q80	"We use accuracy for evaluating logical reasoning performance." (p.6)
Q99	"With few-shot prompting, GPT-3.5 and GPT-4 both perform much better on examples with a 0 ~ 3 reasoning depth, indicating that examples with a 4 ~ 7 reasoning depth pose a challenge to the SoTA LMs." (p.8)
Q155	"Confusion matrices in Figure 4 for the fine-tuning and 8-shot NL prompt results both show that LLMs are significantly better at making the correct predictions for conclusions with labels of True than the conclusions with labels of False or Unknown." (p.15)
Q156	"The accuracy on examples with False or Unknown conclusions is 61.9% with fine-tuning and 54.0% with few-shot prompting." (p.15)
Q157	"They also tend to make more predictions of True than the other labels." (p.15)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality mismatch is total. FOLIO output: single label. Deployment output: numbered Fitch-style proof trace + premise citations + rule names + plain-English summary + (for invalid args) countermodel + register tag.

OF-2: Not applicable — speech/TTS not required for either source or deployment (text-only).

OF-3: Text-only output is accessible for most US undergraduates. Accessibility for screen-reader users with formal-notation output (MathML or equivalent) is a separate platform concern flagged as needing verification.

OF-4: External-validity violation is severe: no FOLIO metric assesses (a) Fitch-style rule-citation correctness, (b) premise-citation correctness, (c) register appropriateness, (d) countermodel validity, (e) pedagogical clarity. The authors themselves note the need for better translation metrics [Q77]. The output-form gap is the deepest validity problem identified.

ID	Evidence
Q77	"We leave for future work the design of a more reliable metric of NL-FOL translation." (p.6)
Q80	"We use accuracy for evaluating logical reasoning performance." (p.6)
Q152	"Although LLMs such as GPT-4 can be utilized to conduct the conversion, it is hard to ensure the GPT-4 outputs are always correct." (p.14)
WEB-7	P-FOLIO adds pass@k over multi-step NL proof chains — closest existing extension but still not Fitch-style rule-cited proofs (https://aclanthology.org/2024.findings-emnlp.966/)
WEB-9	2025 ITS study uses 4-criterion rubric (consistency, clarity, contextual explanation, pedagogical appropriateness) for LLM hints — closest to deployment evaluation but for propositional logic, not Fitch (https://arxiv.org/abs/2505.04736)
WEB-10	No benchmark evaluates formal countermodel construction as first-class output for pedagogical FOL/propositional logic (https://arxiv.org/pdf/2603.19514)

Information Gaps

- Whether any FOLIO-derived rubric for proof-trace quality has been developed in follow-up work beyond P-FOLIO

Requires Expert Verification

- Whether the deployment can use P-FOLIO's NL proof chains as a stepping-stone supplement despite their non-Fitch format

Remediation

Gap 1: Accuracy metric does not evaluate proof-trace quality, register, or countermodel correctness

Recommendation: Develop a multi-component rubric drawing on the 2025 ITS study's criteria [WEB-9] (consistency, clarity, contextual explanation, pedagogical appropriateness) extended with Fitch-specific items (rule-name correctness, premise-citation correctness, subproof-discharge correctness); use both LLM-grader and human-instructor ratings.

ID	Evidence
WEB-9	2025 ITS study uses 4-criterion rubric (consistency, clarity, contextual explanation, pedagogical appropriateness) for LLM hints — closest to deployment evaluation but for propositional logic, not Fitch (https://arxiv.org/abs/2505.04736)